A CASE STUDY IN NUCLEAR ENERGY GOVERNANCE — BANGLADESH

Environmental Impact Assessment of the Rooppur Nuclear Power Plant

Legal framework, baseline environment, impact evaluation and environmental governance of Bangladesh's first nuclear power project, Ishwardi, Pabna District

Prepared for: Academic Assignment / Seminar Submission

Site: Rooppur, Ishwardi Upazila, Pabna District, Rajshahi Division, Bangladesh

Status reference date: 10 August 2026

2×1200 MW

VVER-1200 Units 1 & 2

Prepared by:

Abdulla Al Nahid

ID: 2110756178

CONTENTS

Presentation Outline

Thirteen sections, from legal foundation to environmental governance outcomes

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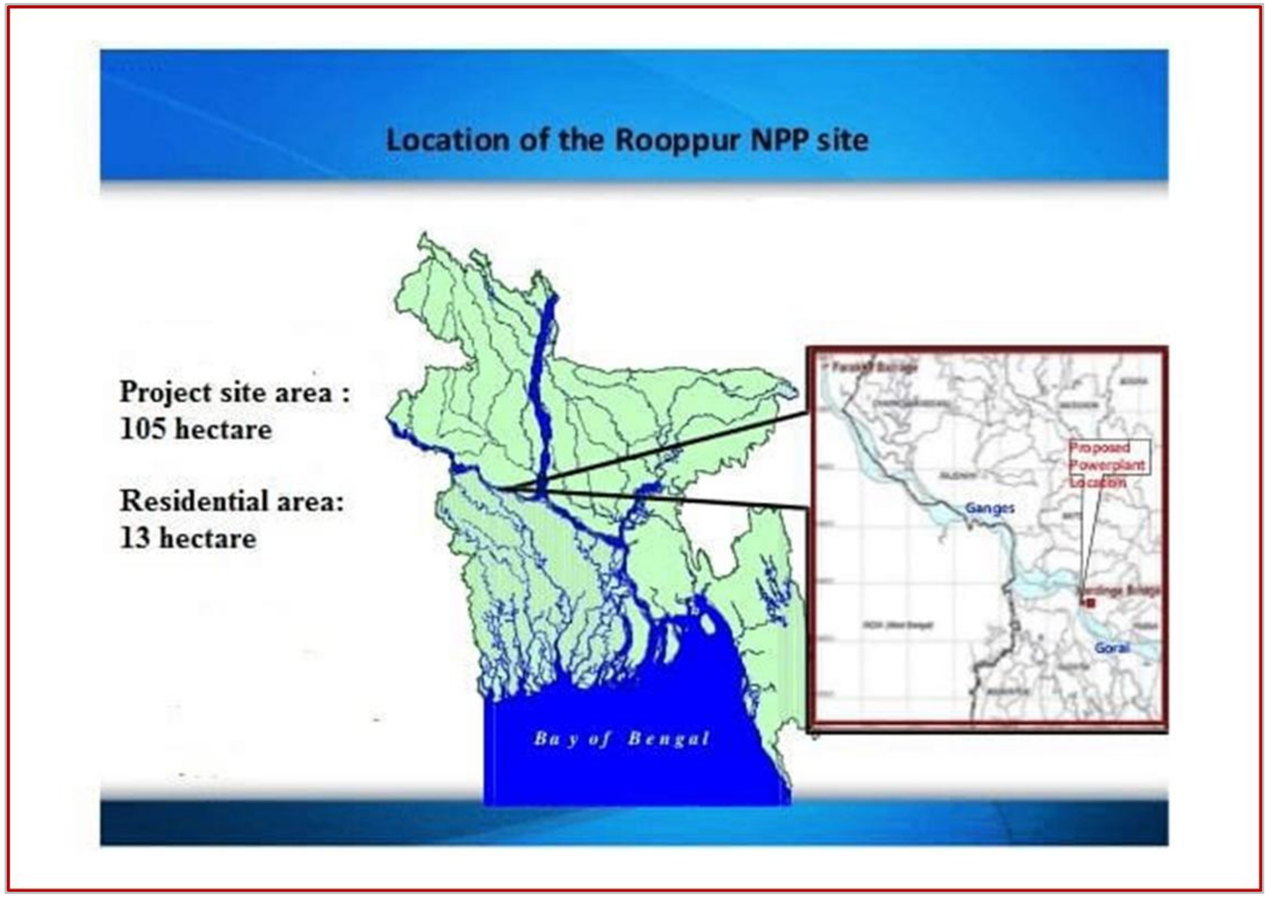
Project Overview & Technical Profile

- Owner / Licensee: Bangladesh Atomic Energy Commission (BAEC), established 1973
- Operator: Nuclear Power Plant Company Bangladesh Ltd. (NPCBL); Rosatom (Russia) as turnkey vendor and first-year operator
- Reactor technology: 2 × VVER-1200 (AES-2006), Generation III+ pressurised water reactors; reference plant Novovoronezh NPP-2
- Financing: State-to-state project; ~90% of the $12.65 billion contract value financed through a Russian state export credit
- Design service life: 50 years (power unit) / 60 years (reactor plant), with technical potential for life extension

Parameter	Value
Thermal output (per unit)	3,200 MWth
Net electrical output (per unit)	~1,150–1,200 MWe
Combined nameplate capacity	up to 2,400 MW
Coolant / cooling source	Padma River (4 natural-draught towers)
Site area	260 acres (105 ha) plant + 32 acres (13 ha) residential
Refuelling interval	12 months

Source: BAERA/FNCA country presentation (2019); World Nuclear Association country profile; Wikipedia, Rooppur NPP.

Site Location & Physical Setting



Location of the Rooppur NPP site on the Padma (Ganges) River, western Bangladesh.

- Location: Rooppur, Ishwardi Upazila, Pabna District, Rajshahi Division — approx. 24.07°N, 89.05°E
- Distance: ~160 km northwest of Dhaka; 20 km east of Pabna town; 8 km south of Ishwardi town
- Hydrological setting: north bank of the Padma (Ganges) River — the plant's cooling-water source
- Site footprint: 105 ha (260 acres) industrial area + 13 ha (32 acres) residential/township area
- Regional hazard context: seismically active zone; river flooding, erosion and cyclone-prone lowland delta

Sources: uploaded BAERA site-location map; radioecological monitoring study, ScienceDirect (2021); ESIA case study, Ishwardi Upazila.

Legal & Regulatory Framework for EIA

ENVIRONMENTAL LEGISLATION

- Bangladesh Environment Conservation Act, 1995 — amended by Acts of 2000, 2002 and 2010
- Environment Conservation Rules (ECR), 1997 — amended seven times, most recently in 2017
- Section 3(1): establishes the Department of Environment (DoE), headed by a Director General
- Section 12(1): no industrial unit may be established without an Environmental Clearance Certificate (ECC)

NUCLEAR-SPECIFIC LEGISLATION

- Bangladesh Atomic Energy Regulatory Act, 2012 — establishes BAERA (formed 12 Feb 2013)
- Nuclear Safety and Radiation Control (NSRC) Rules, 1997
- NSRC Rule 17.3: a Safety Analysis Report, informed by the EIA, underpins the Siting Licence

Regulator / Body	Role in RNPP EIA
Dept. of Environment (DoE)	Reviews and approves the EIA Report; issues ECC
BAERA (est. 2013)	Nuclear regulator; issues Siting and Construction Licences
BAEC (est. 1973)	Project owner/licensee; commissions the EIA
IAEA	Safety standards and international guidance (non-statutory)

Sources: Bangladesh Environment Conservation Act 1995; ECR 1997; BAER Act 2012; NSRC Rules 1997 — as presented by BAERA/FNCA (2019).

EIA Procedure & Environmental Clearance

CLASSIFICATION (ECR '97, RULE 7)

– Industrial projects are classified Green, Orange-A, Orange-B or Red by site and impact

– Nuclear power is explicitly listed as a Red category activity (item 35 of the Red schedule)

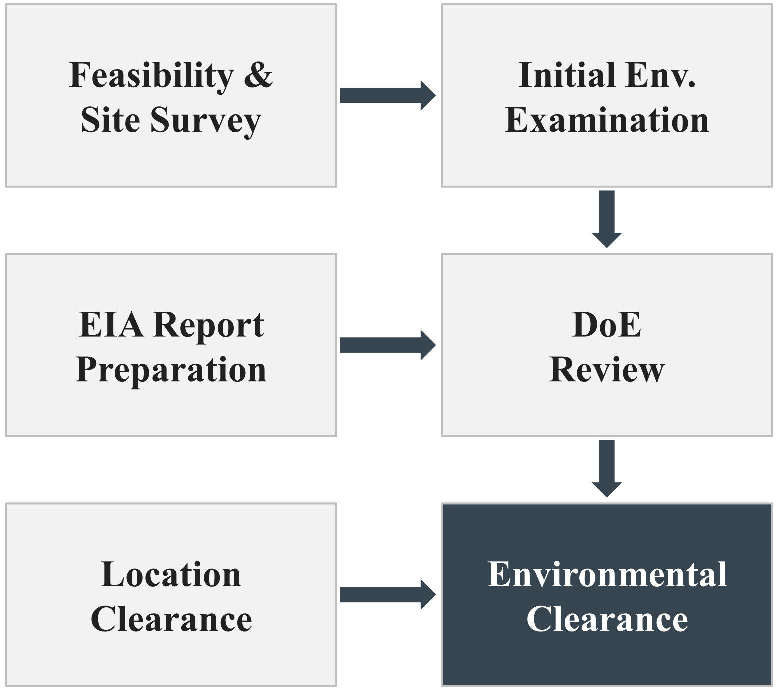
– Red-category projects require sequential clearance: Location Clearance Certificate (LCC) first, then Environmental Clearance Certificate (ECC)

DOCUMENTS REQUIRED FOR ECC (RULE 7.6(D))

– Feasibility report and Initial Environmental Examination (IEE)

– EIA report, Environmental Management Plan (EMP) and process-flow diagram

– No-Objection Certificate from the local authority; emergency and relocation plans



Key milestone: RNPP EIA report submitted to DoE and BAERA; DoE approved the EIA Report on 26 November 2017; BAERA issued the Siting Licence on 21 June 2016.

Sources: Environment Conservation Rules 1997, Rules 7(1),(4),(6)(d); BAERA/FNCA presentation (2019).

Structure & Scope of the RNPP EIA Report

AUTHORSHIP AND APPROVAL

- Prepared by Atomstroyexport (JSC ASE EC) on behalf of BAEC, following Russian and Bangladeshi regulations plus applicable IAEA codes and standards
- Approved by the Department of Environment on 26 November 2017
- Submitted to BAERA in support of the Siting Licence application

AUTHORISATION TIMELINE

Milestone	Date
Siting Licence (BAERA)	21 Jun 2016
EIA Report approved (DoE)	26 Nov 2017
Design & Construction Licence, Unit 1	2 Nov 2017
Design & Construction Licence, Unit 2	8 Jul 2018

CONTENTS OF THE EIA REPORT

- Legal & legislative framework
- Project specification & description
- Land cover / land-use suitability analysis
- Description of the (baseline) environment
- Environmental impacts & their evaluation
- Environmental mitigation & protection measures
- Risk assessment
- Environmental monitoring plan & work plan
- Public consultation / information disclosure

Source: BAERA/FNCA Study Panel presentation, Tokyo, 7 March 2019 — "EIA in the Legal and Regulatory Framework for NPP in Bangladesh."

Baseline Environment I: Physical & Hydrological

HYDROLOGY & LAND USE

- The Padma (Ganges) River is both the cooling-water source and the receiving water body for treated discharge
- Riverbank morphology is dynamic: seasonal char (sandbar) formation, erosion and sedimentation are characteristic of this reach
- Pre-project land use was predominantly agricultural and riverine floodplain, with dispersed rural settlement
- 260 acres (105 ha) were acquired for the industrial area and 32 acres (13 ha) for the residential/township area

SEISMIC & EXTERNAL-HAZARD BASELINE

- Design Basis Earthquake (DBE): 0.172 g | Maximum Design Earthquake (MDE): 0.333 g
- Structures are also designed against aircraft impact (up to 5.7 t at 100 m/s) and extreme internal pressure

Physical parameter	Baseline description
Coolant source	Padma River (surface water)
Cooling towers	4 × natural-draught towers
Topography	Low-lying alluvial floodplain
Regional hazards	Riverine flooding, erosion, seismicity, cyclones
MDE / DBE	0.333 g / 0.172 g

Sources: BAERA/FNCA presentation (2019); World Nuclear Association country profile; radioecological monitoring study (ScienceDirect, 2021).

Baseline Environment II: Pre-Operational Radiological Survey

– IAEA guidance (NG-T-3.11) requires documented environmental monitoring from the pre-construction through post-decommissioning stages of a nuclear facility

– Independent studies (2021–2025) measured naturally occurring radioactive materials (NORM) in soil, sediment and char-land sand around the RNPP site before commissioning

– Findings establish a pre-operational baseline against which future operational monitoring can be compared

– Elevated Th-232 is attributed to local geology (heavy-mineral content) rather than the (not-yet-operating) plant

Medium	Ra-226 (Bq/kg)	Th-232 (Bq/kg)	K-40 (Bq/kg)
Soil (mean)	44.99 ± 3.89	66.28 ± 6.55	553 ± 82
Sediment (mean)	44.59 ± 4.58	67.64 ± 7.93	782 ± 108
Char-land sand (range)	13 – 132	17 – 132	220 – 640

Interpretation

– Most radiological hazard indices fall within globally accepted safe limits

– Some Ra-226 and most Th-232 values exceed the world average — a geological, not operational, signature

– One sand sample showed elevated activity, flagged for continued monitoring

– Sediment geochemistry shows minor-to-moderate contamination by As, Cr and several trace elements, unrelated to plant operation

Sources: Status of NORMs adjacent to RNPP (Sci. Total Environ., 2021); Neutron activation analysis of Padma sediments (2023); Char-land sand radioactivity study (2025).

Baseline Environment III: Socio-Economic Setting

STUDY CONTEXT

- A dedicated Environmental & Social Impact Assessment (ESIA) surveyed a 3 km radius around the RNPP site in Ishwardi Upazila
- Covered settlements include Zhautola, Paksey railway officers' colony, Paper Mill colony, Fisherman colony, Notun Rooppur and neighbouring areas
- Local population depends on the Padma River for drinking, washing, cooking, bathing and fishing — a key receptor pathway for any water-quality impact

LAND ACQUIRED

- 260 acres (105 ha) for the plant + 32 acres (13 ha) for the residential/township area, acquired by the Ministry of Science and Technology

BASELINE HEALTH SURVEY, RNPP WORKERS (CATEGORY-A)

Reported condition	Share of respondents
Respiratory illness	35%
Eye problems	33%
Skin infection	12%
Disease-free	20%

This baseline survey pre-dates commissioning and is not attributed to plant emissions; it establishes a reference point for post-operational occupational health monitoring.

Source: Environmental and Social Impact Assessment (ESIA) of the RNPP surrounding area — case study of Ishwardi Upazila, Bangladesh.

Identified Environmental Impact Categories

Impact category	Source activity	Primary receptor	Assessment priority
Thermal discharge	Once-through & tower cooling water return to Padma River	Aquatic ecology, water temperature	High
Radiological releases	Normal operation & potential accident scenarios	Air, water, public dose, biota	High
Radioactive waste	Spent fuel, low/intermediate-level waste	Long-term storage, transport, security	High
Land use change	105 ha (260-acre) industrial + residential footprint	Agricultural land, settlement pattern	Medium
Hydrology & erosion	Intake/outfall structures, jetty, riverbank works	Padma River morphology	Medium
Air quality	Construction dust, vehicle emissions, standby diesel	Local air quality, nearby settlements	Low–Medium
Socio-economic	Employment, resettlement, in-migration	Local communities, livelihoods	Medium
Ecological baseline	Site clearance, riverine habitat disturbance	Flora, fauna, fish migration	Medium

Impact framework synthesised from the RNPP EIA report contents (DoE-approved, 26 Nov 2017) and the ECR '97 EIA methodology for Red-category projects.

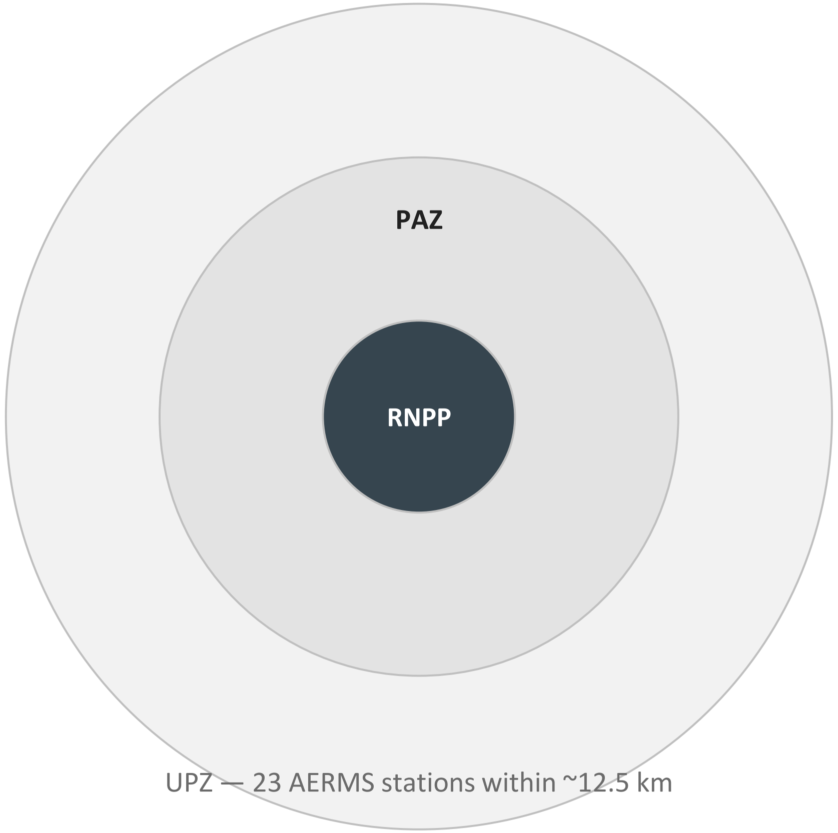
Mitigation Measures & the Environmental Management Plan

Impact	Mitigation measure	Responsible framework
Thermal discharge	Heat dissipation via 4 cooling towers before river return; discharge-temperature limits	Plant operations / EMP monitoring
Radiological release	Five-layer defence-in-depth: fuel cladding, RPV, primary circuit, containment, exclusion zone	BAERA licensing conditions
Severe-accident risk	Core-catcher, double (inner/outer) containment domes, passive safety systems	Design basis (AES-2006 / VVER-1200)
Spent fuel & waste	Cooling-pool storage; classification & treatment; return of spent fuel to Russia under agreement	BAEC / Rosatom fuel-cycle agreement
Land & resettlement	Compensation and resettlement planning for acquired land	Government resettlement framework
Public concern	Nuclear Industry Information Centre; public consultation and disclosure	EIA public-participation requirement

Sources: EIA Guidelines for Industries, DoE (1997); BAERA/FNCA presentation (2019); RNPP defence-in-depth design documentation.

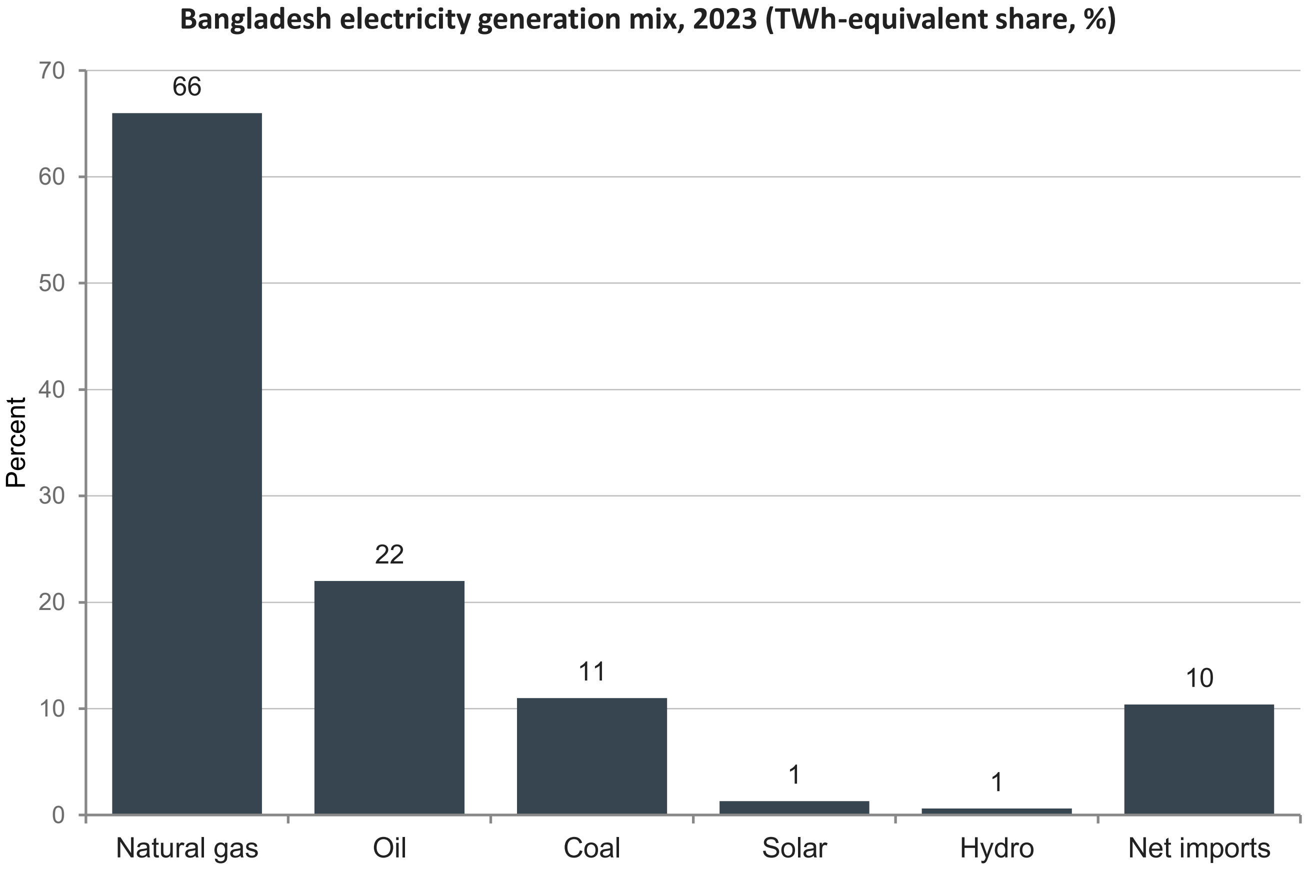
Environmental Monitoring Programme

- An offsite Environmental Radiation Monitoring System (ERMS) covers the Precautionary Action Zone (PAZ) and part of the Urgent Protective Action Zone (UPZ)
- 23 Automated Environmental Radiation Monitoring Stations (AERMS) operate within a 12.5 km radius of the plant
- Station locations include Rooppur High School, Pakshy CNG Filling Station, North Bangla Paper Mill, Bheramara Power Station and the regional cement factory
- Continuous monitoring covers ambient dose rate; periodic sampling covers soil, sediment, water and biota
- Pre-operational baseline studies (NORM, sediment, health survey) provide the reference dataset for detecting future change



Source: RNPP Environmental Radiation Monitoring System description, national thesis on RNPP construction & operation.

Why Nuclear Baseload Matters: Bangladesh's Generation Mix

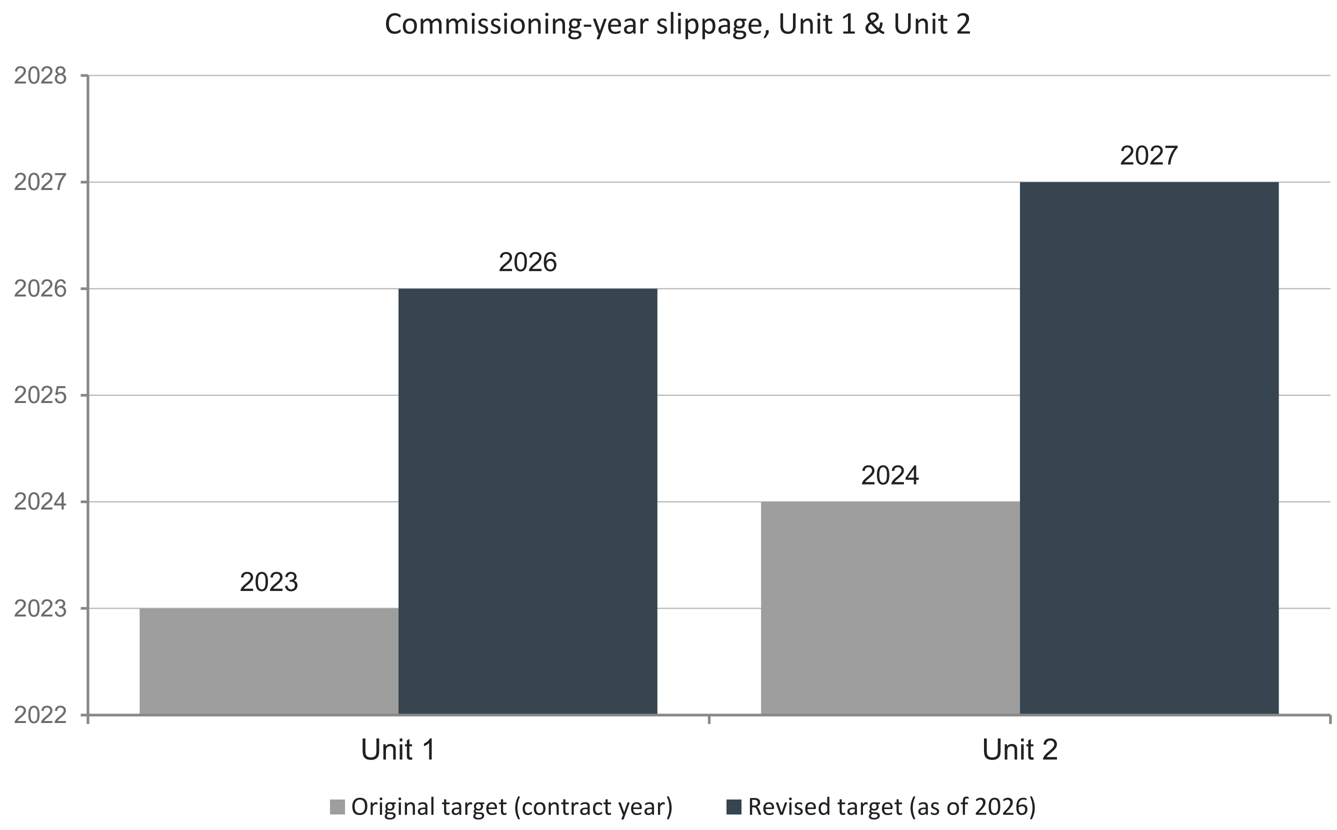


READING THE CHART

- Bangladesh's grid remains overwhelmingly fossil-fuel based: gas, oil and coal supply ~99% of domestic generation
- Demand is growing roughly 7% per year, and roughly 99% of the population already has electricity access
- RNPP's up to 2,400 MW is designed as a carbon-free baseload addition — displacing fossil generation rather than following it

Data: International Energy Agency / World Bank, cited in World Nuclear Association, "Nuclear Power in Bangladesh" country profile (2023 data).

Schedule Risk: Original vs. Revised Commissioning Targets



DRIVERS OF DELAY

- Original contract: Unit 1 by Oct 2023, Unit 2 by Oct 2024
- Delays attributed to the Covid-19 pandemic, the Russia–Ukraine war, currency/financing transfer complications and equipment-logistics constraints
- Currency depreciation alone has added an estimated BDT ~260 billion to total project cost, now near BDT 1.39 trillion
- Overall project completion has been pushed toward mid-2028

Sources: The Business Standard (Oct 2025, Mar 2026); Prothom Alo (Jul 2025, Apr 2026); World Nuclear Association country profile.

Rewards of the RNPP Project

2,400 MW	30,000+	60 yrs	70–80 t
Combined capacity — ~10% of national demand	Jobs generated during the construction phase	Design service life of the reactor plant	Estimated annual uranium fuel requirement

ENERGY SECURITY & ECONOMIC GROWTH

– As a baseload source, RNPP provides continuous power that helps stabilise the western grid zone and reduce load-shedding

– The ADB has estimated that reliable, uninterrupted power could raise manufacturing output materially — relevant for Bangladesh's export-driven economy

ENVIRONMENTAL & TECHNOLOGICAL BENEFITS

– Nuclear generation emits no CO2 during operation, supporting Bangladesh's climate commitments and reducing urban air-pollution pressure

– The project brings technology transfer, workforce training and a new domestic nuclear-safety institution (BAERA)

Sources: BIPSS commentary, "Assessing Risks and Rewards of RNPP" (Zahid); Asian Development Bank estimates as cited therein.

Risk Summary & Trade-Offs

Risk domain	Nature of the concern	Governing / mitigating factor
Nuclear safety	Severe-accident potential; Chernobyl/Fukushima precedent	Defence-in-depth, core catcher, IAEA safety standards
Natural hazards	Cyclones, flooding, seismic activity in the delta setting	Seismic design basis (0.172–0.333 g); flood-resilient siting
Radioactive waste	Waste hazardous for millennia; disposal capacity	Spent-fuel return agreement with Russia; classification/storage protocols
Financing & schedule	Cost overruns, currency depreciation, multi-year delay	State-backed financing; revised contractual milestones
Geopolitical	Dependence on a single foreign vendor (Rosatom/Russia)	Diplomatic engagement; IAEA safeguards; civilian-use framing
Public perception	Historical anxiety around nuclear waste and safety	Public consultation, information centre, transparent monitoring

Source: BIPSS commentary, "Assessing Risks and Rewards of Rooppur Nuclear Power Plant" (Nabib Zahid, bipss.org.bd).

Project Status as of August 2026

- Cold and hot functional tests for Unit 1 were completed during 2025
- Fuel-loading and physical start-up (Stage-B) for Unit 1 began on 28 April 2026
- Commissioning activities under way: fuel installation, boric-acid addition, primary-circuit testing, safety-system verification and sub-criticality testing
- The Ministry (14 June 2026 parliamentary reply) indicated first grid connection around day 114 of fuel loading — approximately end of August 2026 — with an initial output near 300 MW
- Output is planned to ramp in stages (~10% → 40% → 70% → 100%) over roughly 8–10 months to reach full 1,200 MW
- Revised contractual deadlines: Unit 1 by 31 December 2026; Unit 2 by 31 December 2027; overall project completion around mid-2028

EIA relevance of this stage

- First fuel loading and start-up is the point at which operational (not construction-phase) environmental monitoring formally begins
- Pre-operational baselines (radiological, hydrological, health) established through 2017–2025 become the reference for comparison
- Continued transparency in reporting grid-connection milestones is itself part of the public-consultation dimension of the EIA framework

Sources: The Business Standard (14 Jun 2026; 5 Mar 2026); Prothom Alo (29 Apr 2026); World Nuclear Association country profile.

Conclusion & Recommendations

CONCLUSION

- RNPP's EIA followed Bangladesh's Red-category clearance process under ECR '97, supplemented by BAERA's nuclear-specific licensing and IAEA safety guidance
- The DoE-approved EIA report (2017) and subsequent independent baseline studies (2021–2025) together provide a reasonably documented pre-operational environmental record — a genuine strength relative to many first-time nuclear programmes
- The core trade-off is consistent with global nuclear experience: strong climate/air-quality benefits and baseload security, set against long-lived waste-management, cost and schedule risk
- As of August 2026, the project is entering its most environmentally consequential phase — fuel loading and start-up — where monitoring data will begin to be tested against baseline

RECOMMENDATIONS

- Publish operational environmental-monitoring data (AERMS, river-temperature, effluent) on a regular, public schedule
- Commission independent (non-vendor) review of post-commissioning radiological and thermal-discharge results against the pre-operational baseline
- Maintain chain-of-custody transparency for spent-fuel transport under the Russia return agreement
- Continue capacity-building for Bangladeshi environmental and nuclear-safety specialists at BAERA and NPCBL
- Extend socio-economic and occupational-health monitoring beyond the construction-phase baseline into steady-state operation

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– The Bangladesh Environment Conservation Act, 1995; Environment Conservation Rules, 1997; Bangladesh Atomic Energy Regulatory Act, 2012; Nuclear Safety and Radiation Control Rules, 1997.

– Wikipedia, "Rooppur Nuclear Power Plant" (infobox data, cross-checked against WNA).